

Python Automata Library (PAL) Cheat Sheet

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import PythonAutomataLibrary as pal.  
Safe mode is default. Call pal.FastMode() before  
constructing any PAL object. Put substantive model  
kernels under @pal.njit(cache=True).
```

Create state

pal.NewGrid(dims, dtype) — typed lattice
pal.NewAgentGrid(dims, numAgentProps=0, isStackable=False) — individual agents;
dims=() is nonspatial
pal.NewPopGrid(dims, capacity=None) — integer population counts
pal.NewPDEgrid(dims) — continuous field
pal.NewIList() · **pal.NewMultinomial()** — integer query list · random-count sampler

Dimensions — 1–3 axes; a negative dimension wraps that axis, e.g. (-nx, ny).

Shared lattice geometry

Properties — xDim/yDim/zDim, nDims, wrapX/Y/Z

Index conversion — ToI(x[,y,z]); ItoX/Y/Z(i)

Regions — Box(lo..., hi...) uses half-open bounds; Hood(hood, x[,y,z]) maps relative offsets to coordinates.

Neighborhoods — pal.MooreHood(dim, includeOrigin) · pal.VonNeumannHood(dim) · pal.CircleHood(dim, rad)

Grid / common indexing

Read/write — g[x,y], g[x,y]=v; slices return detached NumPy copies.

Pattern — g = pal.NewGrid((40,40), float) · g[10,12] = 1.0 · v = g[10,12]

AgentGrid

Create / move — NewAgentSQ(x,y) / MoveSQ(a,x,y) for lattice positions; NewAgent(x,y) / Move(a,x,y) for continuous positions.

Agent state — grid[a,p] property; I(a), XSQ/YSQ/ ZSQ(a) lattice; X/Y/Z(a) continuous; Alive(a), Dispose(a).

Queries — GetPop(), AgentsAt(...), LastAgent(...), AgentsInRadius(...), counts[...].

Iteration — for a in agents.All(): ... is a snapshot and is safe for structural mutation such as Dispose.

Wrapping — DispWrapX/Y/Z(p1,p2) gives wrapped displacement.

PopGrid + PDEgrid

Transactional update — Add(v, ...) changes pending state; Update() applies it simultaneously; Reset() clears current + pending. Direct grid[...] = v changes current state immediately.

Population — pop.GetPop() total; pop.All() nonzero site indices. capacity limits total population.

PDE setup — field.SetTimeSpaceStep(dt, dx[,dy,dz])

Diffusion — Diffusion, DiffusionMask, DiffusionField, DiffusionInterfaces, DiffusionADI; radial 1D: DiffusionRadialCircle/Sphere.

Advection — Advection, AdvectionField, AdvectionInterfaces.

Lists + randomness

IList — Append(i), Clear(), Random(), Shuffle(), indexing/len; All() detached copy; Iter() no-copy iteration.

RNG — pal.Seed(seed) · pal.Random() · pal.RandInt(n) → 0..n-1; use PAL RNG for shared-stream reproducibility.

Multinomial — m.Binomial(n,p); Setup(...) then Sample(...) for repeated multinomial draws.

Draw / output

Pixels — pix, win = pal.StartPixWindow(xDim,yDim,scale=1,title='PAL',headless=False); set pix[x,y]=RGB; win.Update(); win.Save(path, block=True); win.Close().

GIF — StartGif(path,delay=100) · AddGifFrame(block=False) · StopGif(); call Update() before capture.

OpenGL — pal.StartOpenGLWindow(...); draw with Circle, Box, BoxSQ, Line, Borders; scene controls include Camera, Background, Clear.

More detail: Manual · API Guide · API Reference (Documentation/).